Farmers' requirements for sorghum seed products in Tanzania

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Abstract

This brief explores Tanzanian farmers' requirements for sorghum seed products as input for discussions by CGIAR and NARES on seed product market segmentation and the design of target product profiles (TPPs). We applied a novel approach to identify the requirements—video-based product concept testing (VPCT). Through multiple engagements with breeders, farmers, and industry, we identified seven sorghum seed-product concepts, five targeting the end use—home use, feed and food, industrial malting, food and fodder, forage; one concept on cropping system—intercropping; and one targeting material type—hybrid. We sampled 1,100 farmers from the Dodoma and Shinyanga Regions, where each farmer evaluated and ranked three concepts. Farmers were most likely to have selected the hybrid concept as their most preferred concept, followed by the home use concept. Based on these results, we present a proposal for revisions of the current set of market segments and TPPs for sorghum in Tanzania—one the largest sorghum producers in Eastern Africa.

Key Points

- Among the seven sorghum seed-product concepts developed (home use, feed and food, industrial malting, food and fodder, forage, intercropping, and hybrid), farmers were most likely to have selected hybrid and home use as their preferred concepts. The intercropping concept ranked third, followed by feed and food, food and fodder, industrial malting, and forage.
- The preference for the hybrid concept reflects the growing interest in sorghum as a cash crop for the brewing industry and export. Regional exports of sorghum from Tanzania have increased by 300 percent over the past five years.
- The high priority given to the home use concept, which would result in a 20 percent lower yield, considers that sorghum forms an important part of rural diets and that its value is in home use. Future conversations with farmers, processors, and food scientists are required to better understand what traits determine "good taste" and how best to measure these traits.
- Across all the concepts, white sorghum grain was preferred by 70 percent of the sample. White color is a must-have requirement for national and regional markets and is also considered to provide better tasting sorghum-based foods.
- The results support discussions on refinements to sorghum segmentation. The hybrid segment should be prioritized, while the following new segments should be considered: home use, intercropping, food and fodder, food and feed. The forage segment should be maintained as a future market segment.

INITIATIVE ON Market Intelligence

CGIAR

This Initiative aims to maximize CGIAR and partners' returns on investment in breeding, seed systems and other Initiatives based on reliable and timely market intelligence that enables stronger demand orientation and strengthens co-ownership and co-implementation by CGIAR and partners.

Introduction

Across Eastern and Southern Africa, the demand for sorghum is growing. Orr et al. (2020) projected that the demand for sorghum in Eastern and Southern Africa would reach 29.9 million t by 2050 under a "business as usual" scenario, up from 11.8 million in 2015 (an increase of 153 percent). Sorghum production is concentrated in the region's semiarid agroecologies where farmers regard them as insurance crops and are unwilling to rely solely on maize for household food security. Consumers perceive sorghum production as complementary to maize production. Sorghum can also be mixed with wheat flour to produce composite wheat flour (Poole et al. 2024). The market is growing for sorghum grain for animal feed and industrial brewing, even in countries where sorghum is traditionally grown for food (Srinivasa Rao et al. 2014). Population growth, consumer preferences, and growth in brewing and livestock-feed industries are driving demand up for sorghum across Africa. Against this background, future investments in sorghum breeding and seed systems development by CGIAR and the national agricultural research and extension Systems (NARES) have relatively high potential to deliver impact on economic and food security.

Tanzania is one the largest sorghum producers in Eastern and Southern Africa, and the seventh largest producer globally. In area under cultivation, sorghum is the third most grown cereal in Tanzania, after maize and rice. It is mainly grown in semiarid regions under rainfed production systems. Most farmers intercrop sorghum with other crops, including legumes, maize, and cassava (Slakie et al. 2019). In the past decade, sorghum production has grown substantially. The quantity of sorghum produced increased by about 20 percent (Figure 1), from 838,717 t in 2012 to 1 million t in 2022 (FAOSTAT 2023). In addition, the area harvested increased by a similar percentage, from 839,423 ha in 2012 to 1 million ha in 2022. A growing export market has played a role in the expansion of sorghum production. The value of exported sorghum increased from USD 2.4 million in 2012 to almost USD 11.4 in 2022, which is more than a 300 percent increase in the past five years (FAOSAT 2023).

Traditionally, sorghum has been a staple crop in Tanzania, forming an important component of rural diets. Most farmers produce sorghum primarily for their own consumption, where it is eaten as porridge or ugali (thick porridge), a traditional dish derived from mixing sorghum flour or a mixture of sorghum and maize or cassava flours with water. Portions of the sorghum harvest may also be used as feed for livestock. In past years, increased domestic demand has arisen for white sorghum from breweries such as Tanzania Breweries Company Limited and Serengeti Breweries Limited (Akpo et al. 2022a). The share of domestic use was estimated by Ndossi et al. (2021) as follows: brewing industry (43 percent), food (40 percent), and animal feed (7 percent). Traders (small and large scale) and processors are the main buyers in the sorghum grain market (Akpo et al. 2022a). White and red sorghum are available for home consumption and animal feed.

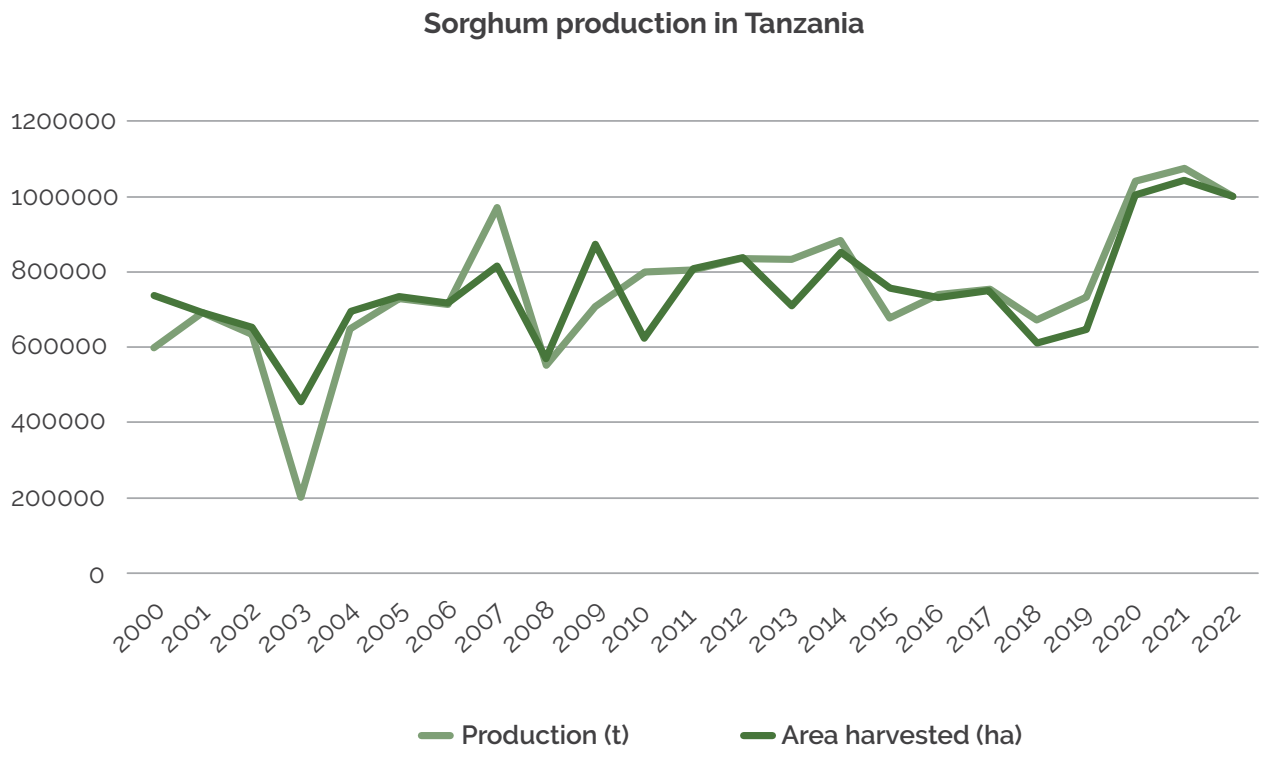


Figure 1. Sorghum production and harvested area in Tanzania (FAO 2023).

Sorghum seed product design and breeding priorities

Research and development efforts focused on sorghum in Tanzania began in the early 1980s. Since that time, international research institutes such as the International Crops Research Institute for Semi-Arid Tropics (ICRISAT), along with national agricultural research systems, notably the Tanzania Agricultural Research Institute (TARI), and various universities have worked together to develop improved sorghum varieties. According to the Tanzania Official Seed Certification institute (TOSCI), 2024, 18 improved varieties were released in Tanzania by 2023. Most of the improved cultivars are designed based on conventional breeding approaches, with a primary focus on agronomic traits, particularly yield and disease resistance. However, the efforts and investments targeted at enhancing uptake of improved varieties have only been partially successful, with significantly low adoption rates (Kimbi et al. 2020). One of the reasons for the low rate is development of products that do not respond to the requirements of farmers and end users (processors and consumers).

Moving forward, the One CGIAR Initiative on Market Intelligence has an opportunity to inform crop-breeding efforts of CGIAR and NARES by shedding light on the requirements for farmers, processors, and consumers and the ideal seed products to respond to these requirements. The two basic elements to guide discussions are seed product market segments (SPMSs) and target product profiles (TPPs). The SPMS refers to a group of farmers with common variety requirements. In the context of CGIAR and NARES crop breeding, these segments are defined based on grower requirements (where and how the crops are grown) and end-user requirements (what the crop is used for). For each SPMS, there is a unique TPP that provides the blueprint for design of the ideal product for that SPMS and indicates the traits and characteristics required for a new variety to meet or exceed the requirements of growers, processors, and consumers.

Table 1 presents the SPMSs for sorghum in the Eastern African Federation (EAF). These segments evolved out of discussions with the Tanzania Sorghum Product Design Team, a group of representatives from public and private sectors that informs SPMS and TPP design, that include farmer representatives, processors, seed companies, as well as scientists such as breeders, pathologists, seed systems specialists, and socio-economists. The main criteria that distinguish the sorghum segments are material type (hybrid versus open-pollinated variety, or OPV) and color (white/cream versus red/brown). The market segment for forage sorghum that is produced for fodder only is listed as "not current country priority" and therefore is not included as a market segment.

All the sorghum SPMSs include a variety of uses: food, feed, fodder, and malting. The corresponding TPPs for these segments contain 29 essential traits (i.e., targets that must be met for the advancement of new varieties), of which 3 to 4 are improvement traits. Delivering new products that meet the thresholds and targets of the 29 traits will be a significant undertaking. Production and agronomic and disease traits such as genetic purity, plant height, disease and pest tolerance, drought tolerance, and early maturity are required for all the seed products. However, there are some nutrition-enhancement and processing traits (such as proteins, starch, iron, and zinc contents; taste; grain color; tannin levels; and digestibility) that are linked to specific use (Table 2). By understanding farmer use requirements, a prioritization of end-use improvement traits can help to simplify the breeding process.

Table 1. Market segments for sorghum in Tanzania

	Market Segment 1	Market Segment 2	Market Segment 3
Material type	OPV	OPV	Hybrid
Subregion	EAF	EAF	EAF
End use	Food, feed, fodder malting	Food, feed, fodder, malting	Food, feed, fodder, malting
Color (grain)	White/cream	Red/brown	White/cream
Production environment	Semiarid lowland/ subhumid midland	Semiarid lowland/subhumid midland	Semiarid lowland/ subhumid midland
Production system	Rainfed	Rainfed	Rainfed
Maturity	Early to medium	Early to medium	Early to medium
Size (ha)	561,989	312,216	8742
Proportion of investment in 2023-2024	60%	20%	20%

Note: OPV = open-pollinated variety; EAF = Eastern African Federation.

Table 2. Sorghum traits important to different end uses

Trait	End Use			
	Food	Feed	Fodder	Malting
Grain color	✓	✓		✓
Starch extract				✓
Protein content	✓	✓		
Stay green until maturity			✓	
In vitro organic matter digestability (IVOM)			✓	
Food taste	✓			
Iron content	✓			
Zinc content	✓			
Tannins	✓	✓		✓

A quick review of the sorghum market segments leads to several questions for discussion with the Tanzania product design team (PDT), related to farmers' requirements for sorghum seed products:

- End use: Do farmers require products that meet a host of possible end uses (e.g., food, feed, fodder, and malting) or are their requirements more focused on one or two end uses. The latter would allow for more feasible TPPs with few essential traits.
- Color: In what color do farmers require the sorghum for different uses? Prioritization of different uses should include color considerations.
- Production systems: TPPs are designed considering a monoculture production system. Do farmers grow sorghum under monoculture or intercropping production systems? If the latter, then this would imply the need to add a market segment that recognizes the intercropping production systems.

Methods

This study used video-based product concept testing (VPCT). Product concept testing is early-stage market research that involves description of hypothetical products, and their potential uses and benefits to the customers. Product concepts (ideas) are usually developed, packaged, and presented to the consumers for evaluation. The underlying principle is that customers seek value from a product rather than the physical product itself (Luso 1975). For example, farmers will experiment with or invest in a new seed product only if they expect the product will help accomplish a particular task or goal (Campos 2021). In some cases, this goal may be higher yield under monocropping conditions; in other cases, the goal may be better-tasting food for the family and local market. Understanding the goals farmers seek from seed products will allow for nuanced segmentation and, in return, more impactful TPPs. In this brief, concept testing focuses on understanding the goals farmers have from producing sorghum and linking these goals to potential future seed products. Sorghum product concepts were designed mainly around specific farmer requirements related to end-user requirements to identify alternatives to the current sorghum SPMS for TPP prioritization in Tanzania. Additionally, a concept of cropping system and the material type was included.

Ideation of sorghum seed product concepts

The process of concept development involved qualitative research, including eight focus-group discussions with smallholders and interviews with various stakeholders, such as seed producers, traders, processors, and TARI, a national plant-breeding institute. Participation in a sorghum PDT meeting in Tanzania and continuous consultations with sorghum breeders at the International Maize and Wheat Improvement Center (CIMMYT) were also integral to the concept development. The discussions focused on the goals farmers had for their sorghum production, their preferred growing methods, uses of sorghum at home, experiences in selling sorghum, and preferences for maturity and color. Discussions with technical experts focused on the technological possibilities of breeding; the new types of products end users (processors and farmers) were interested in; and the suitability and potential of other types of sorghum not yet grown in Tanzania, such as hybrid sorghum and intercropping sorghum. Figure 2 shows the process of ideation of the sorghum concepts.

Seven product concepts emerged from the ideation process (Table 3): Home use (food for the family), feed and food, industrial malting (grain for beer), forage, food and fodder, intercropping (sorghum and peas), and commercial (high-yielding hybrid) products. See Annex 1 for a full overview of the concept's description. The main growers' goals of early maturity, high yield (compared to local varieties), and drought and disease tolerance were held constant in all the developed concepts. Farmers also indicated the color they would require in the products. The description of the concepts included the product goal(s) and any potential shortcomings that the use of the product may imply (e.g., reduced yield).

The product-presentation videos were recorded in a professional studio in the local language (Swahili). A male actor posed as an agrodealer representative, dressed in a white lab coat standing behind a counter and in front of several bags of different seeds, herbicides, and pesticides.

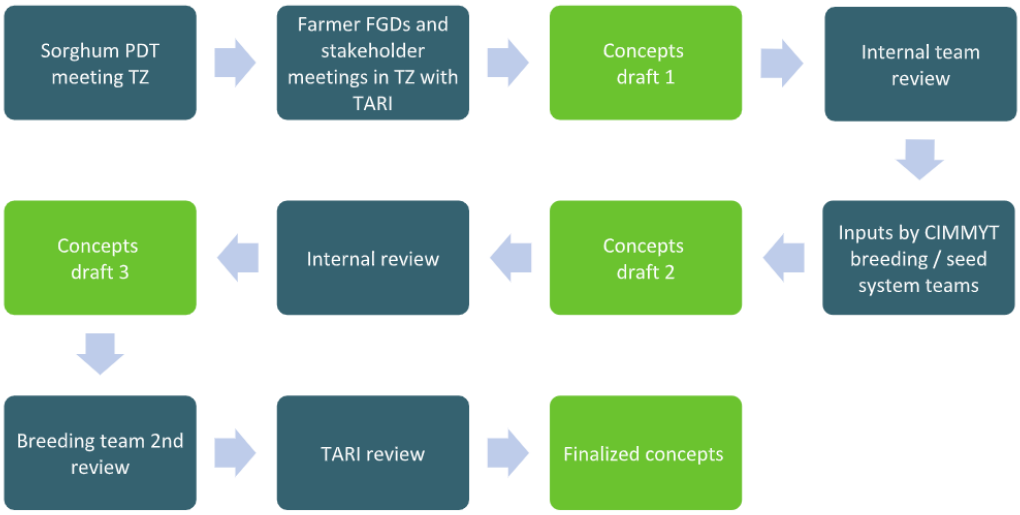


Figure 2. The process of ideation of sorghum concepts in Tanzania.

Table 3. Sorghum seed product concepts

Product	Description of grower or end use goals
Home use product: Food for the family	• Designed for food—nutrition, high flour conversion rates, good taste, cooks easily, easy to thresh • 20 percent less yield compared with other improved seeds • Not for animal feed, fodder, or malting
Feed and food product	• Improved protein for feed • Good for food but less attention has been given to cooking qualities such as taste • Not for fodder or malting
Industrial malting product: Grain for beer	• For processing for industrial beer making • For income purposes only • Not designed for food, feed, or fodder
Forage product: Forage for animals	• Has soft leaves and stems easy for the animals to eat and get nutrients • No grains
Dual purpose product: Food and fodder	• Has big and soft leaves and stems easily eaten by livestock for fodder • Stay green until maturity • Used for food but less attention has been given to cooking qualities • Reduced grain yield of the plant • Not designed for malting and grains for animal feed
Intercropping product: Sorghum and peas	• Performs well when intercropped with legumes • Improves soil and yields and food diversification • Increased cost of management and harvest
Commercial sorghum: High-yielding hybrid	• 30 percent increase in yields • Used for food but less attention has been given to cooking qualities • Buy seeds every season • Not for feeding and malting

A total of nine videos were recorded, a video for each of the seven concepts plus an introductory and a closing video. In the product-presentation videos, lasting one to two minutes, the product was introduced and then the benefits and drawbacks of the concept were highlighted. The introductory video presented the scene (context) and explained what would follow, including the decision that the farmer would need to make. The closing video included guidance on the ranking activity and appreciation for listening.

To evaluate the concepts, a household survey, which included 1,100 sorghum farmers, was carried out in Kongwa and Kondoa Districts of the Dodoma Region and Kishapu and Shinyanga Districts of the Shinyanga Region. Inspired by the tricot approach that uses incomplete randomized blocks, each respondent watched three different product concepts videos (out of the seven). The videos were randomized in subsets of three, where each concept was shown an equal number of times. In total, each farmer watched five videos: the introductory video, three videos describing different product concepts, and a closing video. After the respondents watched each concept video, they answered questions on each concept's similarity to the varieties they currently grow, their interest in trying the new concepts, their willingness to replace their current varieties with the new ones, and the color they would prefer for the products. Finally, respondents were asked to select their most and least preferred product for purchase if available at local agrodealer shops. The data analysis on the ranking of the concept was done by ClimMob software (de Sousa et al. 2022; Turner et al. 2020).

Results of concept testing

The seven product concepts were assessed by ranking them across the whole sample (Figure 3). Table 4 shows the direct rankings among all the concepts, indicating the percentage of farmers who ranked a given concept (indicated in the row) before the other respective concepts (indicated in the columns). Additionally, we used regions, districts, and gender as covariates to split the sample in case of significant instability in the Plackett-Luce model, driven by difference in rankings (Figure 4). Figure 5 indicates the color preference for each concept. We discuss the seven concepts in categories following the farmer ranking: the first tier, the middle, and the last tier, given their probability of being ranked as the most preferred, respectively.

The best performing concepts (first tier) were the high-yielding hybrid and food for the family (Figure 3). They had the highest probability of being ranked as the most preferred by the farmers. In the direct ranking between the concepts (Table 4), the two concepts had a higher percentage of farmers preferring them over the other five concepts. However, in a direct comparison with each other, high-yielding hybrid had a slightly higher win probability compared with the food-for-the-family concept. The Plackett-Luce model (Figure 4), however, indicated that there was some divergence between regions, districts, and gender. Women in Dodoma (in both Kongwa and Kondoa Districts) and men in Kongwa District preferred the concept of food for the family over the high-yielding hybrid, while in the Shinyanga Region (Kishapu and Shinyanga Districts) both men and women preferred the concept of high-yielding hybrid to that of food for the family.

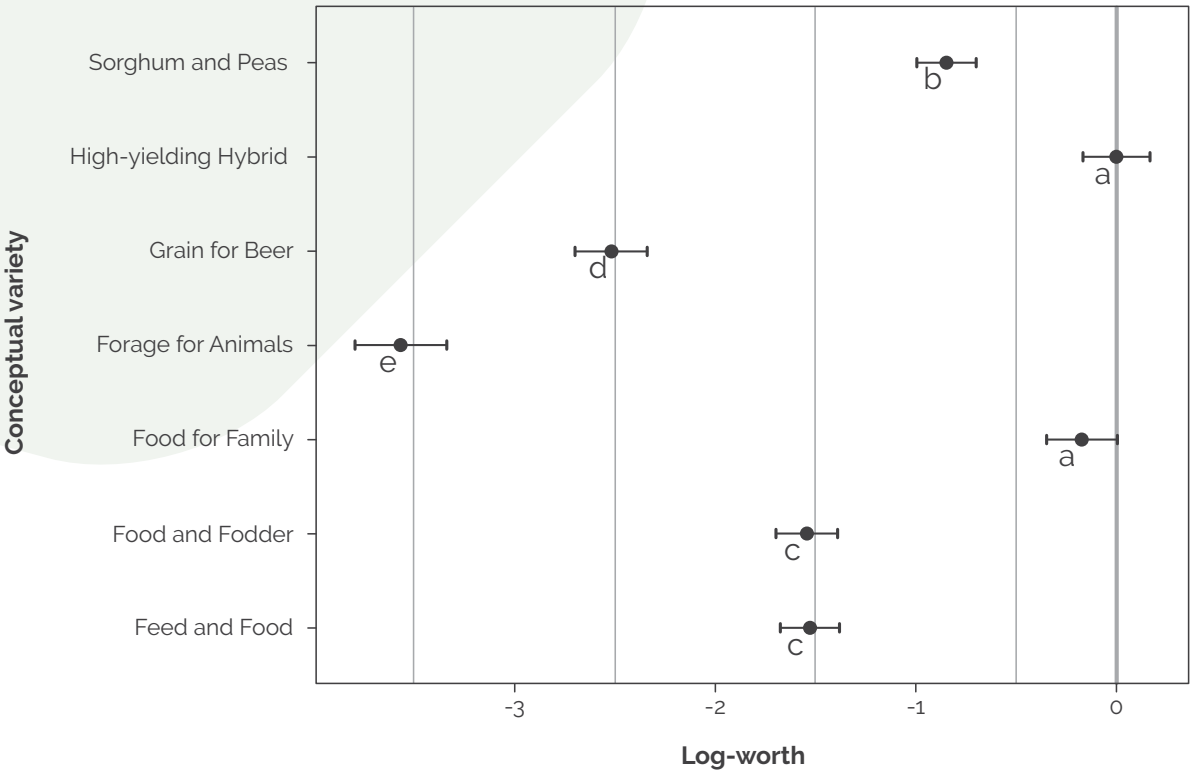


Figure 3. Overall acceptability (log-worth) of product concepts of sorghum presented to farmers in Tanzania. *Log-worths are model coefficients derived from the Plackett-Luce model, which estimates the probability of one given concept in outperforming all the other concepts in the set. The concept* "High-yielding hybrid" *(benchmark) is set as a reference (log-worth arbitrarily set to zero). Different letters indicate significant differences at $p < 0.05$.*

The middle tier includes the intercropping (sorghum and pea), food and fodder, and food and feed (Figure 3). The intercropping concept was the most preferred of the three concepts, while there was no clear leader between food and feed and food and fodder (Table 3). However, men in the Kondoa District, Shinyanga Region, preferred the food and feed concept slightly more than the intercropping concept (Figure 4). The last tier includes concepts that were least preferred, namely grain for beer and sorghum for forage (Figure 3). These concepts were not prioritized by the farmers, especially sorghum for forage, which scored lower than the rest of the concepts. However, men in the Kondoa District, Shinyanga Region, preferred the forage concept slightly more than the grain for beer (Figure 4). Across all the concepts, white sorghum grain was preferred by 70 percent of the sample (Figure 5).

Table 4. Win percentage of each concept in direct comparison with every other concept

	Sorghum and Peas (%)	High-Yielding Hybrid (%)	Grain for Beer (%)	Forage for Animals (%)	Food for Family (%)	Food and Fodder (%)	Feed and Food (%)
Sorghum and Peas	-	30	84	94	34	67	66
High-yielding Hybrid	70	-	93	97	54	82	82
Grain for Beer	16	7	-	74	9	27	27
Forage for Animals	6	3	26	-	3	12	12
Food for Family	66	46		97	-	80	79
Food and Fodder	33	18	73	88	20	-	50
Feed and Food	34	18	73	88	21	50	-

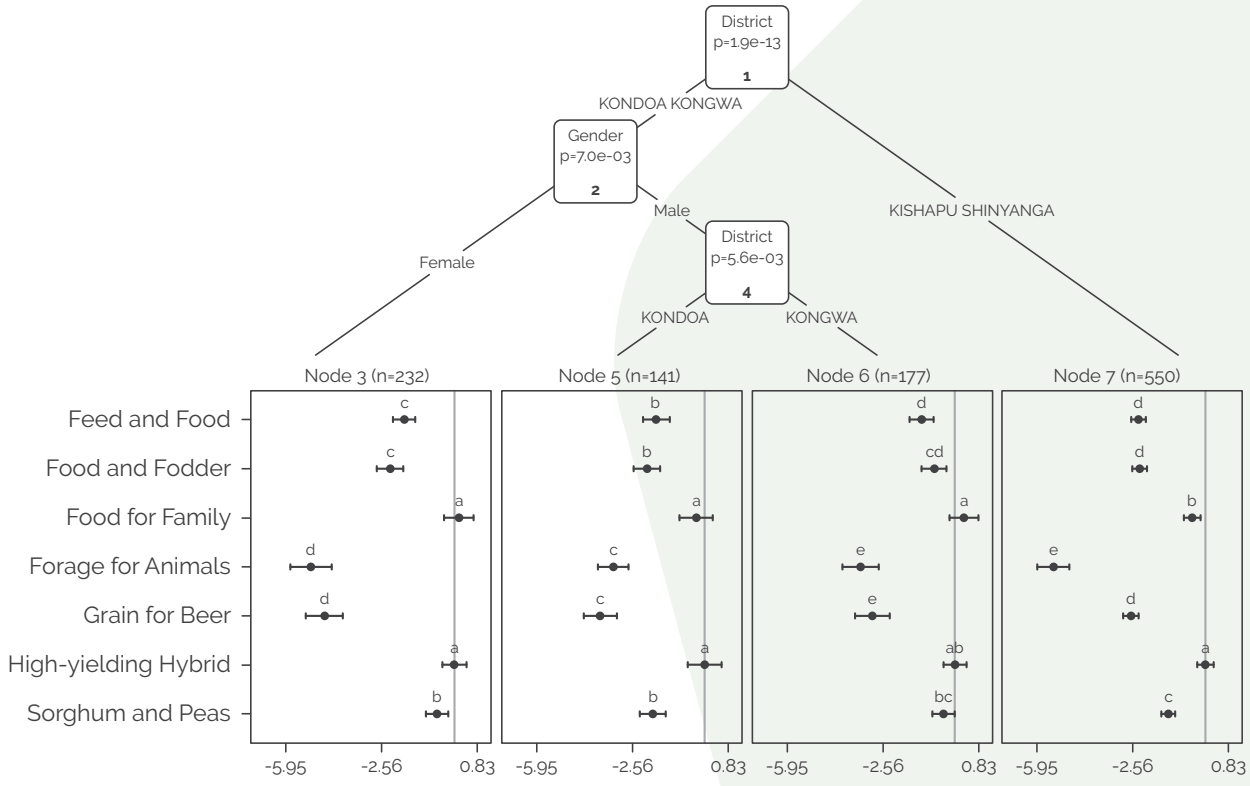


Figure 4. Plackett–Luce Tree of product concept data for male and female participants Different districts in Tanzania. *The horizontal axis of each panel shows coefficients derived from the Plackett-Luce model (log-worth), which estimates the probability of one given concept in outperforming all the other concepts in the set. Error bars show quasi-SEs. The gray vertical lines indicate the reference concept "High-yielding hybrid" (benchmark) (log-worth set to zero). Different letters indicate significant differences at p < 0.05.*

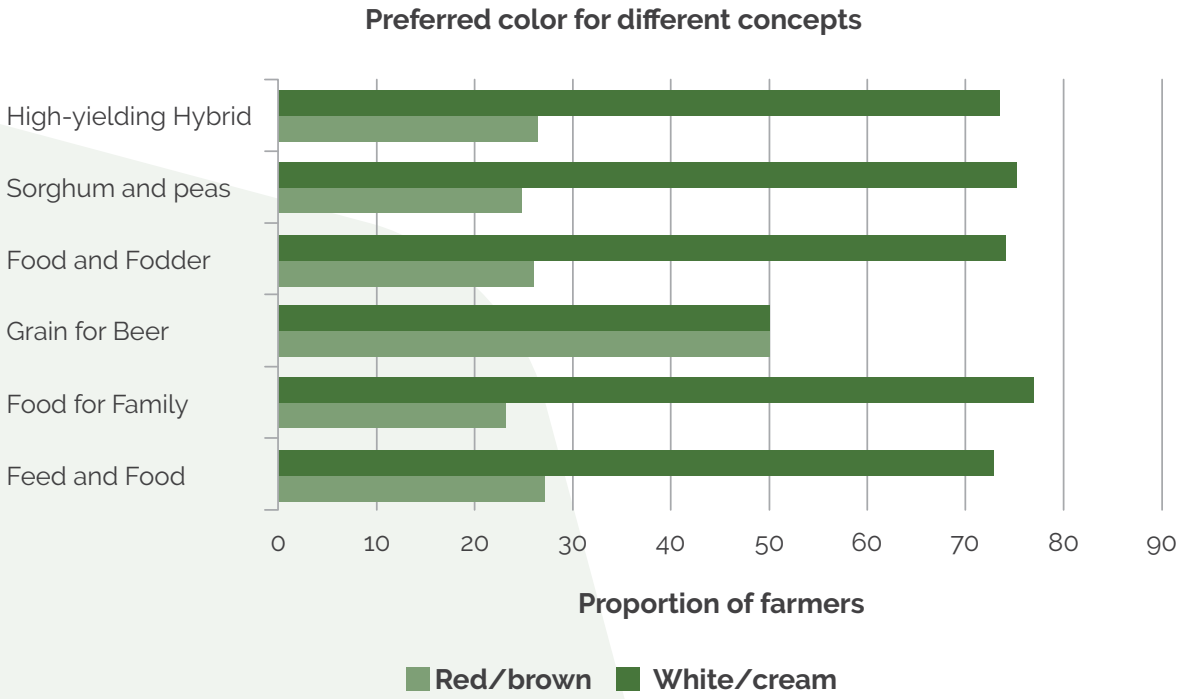


Figure 5. Preferred color for each concept.

Discussion and way forward

Among the seven sorghum-seed product concepts developed, farmers were most likely to have selected hybrid and home use concepts as their preferred concepts. The intercropping concept ranked third, followed by feed and food, food and fodder, industrial malting, and forage. The preference for the hybrid concept indicates a rising interest in sorghum as a cash crop for the brewing industry and for export. Over the past five years, regional exports of sorghum from Tanzania have surged by 300 percent (FAOSTAT 2023). For the concept on food for the family, farmers were willing to sacrifice 20 percent of yield for a product that has good home consumption qualities, which highlights that sorghum forms an important part of rural diets and that its value is in home use. The preference for the intercropping product concept reflects farmers' interest in sorghum seed products that complement legume production. Sorghum for forage was the least preferred concept, indicating that the value of sorghum is largely for grain production. Across all the concepts, white sorghum grain was preferred by 70 percent of the sample. White color is a must-have requirement for national and regional markets and is also considered to provide better-tasting sorghum-based foods.

Refining of market segments

The results support discussions on refinements to sorghum segmentation. First, the study identified hybrid and home use (food for the family) as priority market segments. There is an existing breeding pipeline for hybrids in Tanzania that supplies 2 percent of the market. The strong preference for hybrids indicates the need for increased investments in breeding and seed systems of hybrid seed products. A new current market segment was identified—the home use segment. Conversations with farmers, processors, and food scientists are required to better understand what traits determine "good taste" and "good cooking qualities" and how best to measure these traits.

Second, three near-term future market segments were identified: intercropping (sorghum and peas), food and feed, and food and fodder. Currently, breeding efforts do not support the intercropping segment, but it is a recognized need among farmers. Third, a potentially long-term future market segment was recognized: forage sorghum. Currently, these seed products are unavailable in Tanzania and are not in demand among farmers. Nevertheless, this does not mean that they should be overlooked, especially with increased climate variability in mind: forage sorghum is outstandingly drought resistant and grows where maize is not able to grow because of high temperatures or dry conditions. In future it could be a good substitute for maize in silage making for livestock. Finally, white sorghum varieties should be prioritized.

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Annex 1. Sorghum concept description and video scripts

Product	Concept description/Video Script
Feed and food product	This product is called "Feed and Food." This sorghum seed was designed to provide your family with grain that is useful for feed and food. The seed provides grain with improved protein, which makes it suitable as feed for your own chickens and pigs. You could also sell the grain to other farmers or traders who process animal feed. The seed is also used for food, but less attention has been given to cooking qualities such as taste. The seed may not be as good for food as a product that is designed for food only. The seed is not designed for fodder or malting.
Home use product	This product is called "Food for family." The sorghum seed is focused on serving your cooking needs; the cooking qualities has been given priority. • Nutrition: higher level of iron and zinc • The grain gives a good amount of flour when milled. • The product gives a sweet taste to your uji or ugali. • It cooks easily. • It is easy to thresh. The seed has 20% less yield compared to the other improved seed. This means that if the other improved seed give you 10 bags, this seed will give you 8 bags assuming you planted on a similar farm. However, it still has higher yield than the local varieties. The seed is not designed for animal feed, fodder, or malting.
Industrial malting product	This product is called "Grain for beer." This sorghum seed is mainly focused on serving the needs of the processors in your area. It has large grains and is highly demanded in your area by traders and processors for industrial beer making. This is a sorghum seed for income purposes only. It is not designed for food, feed, or fodder.

Annex 1. Sorghum concept description and video scripts

Product	Concept description/Video Script
Forage product	This product is called "Forage for animals." It is a hybrid sorghum seed designed for making livestock feed. It is harvested before the grain matures, at 100–120 days. It has soft leaves and stems that are easy for the animals to eat and get nutrients. You can use it to make silage for your livestock or sell the silage for income. This seed is not recommended if you grow sorghum for grains.
Dual purpose: Food/fodder	This product is called "Food and fodder." The sorghum seed is designed with qualities that make it suitable for food and fodder. It has big and soft leaves and stems that are easily eaten by livestock for fodder. However, this quality reduces the grain yield of the plant. The seed is also used for food, but less attention has been given to cooking qualities such as taste. The seed may not be as good as for a product that is designed for food only. The seed is not designed for malting and grains for animal feed.
Intercropping variety	This product is called "Sorghum and peas." This sorghum seed is designed to perform well when you combine it in the field with peas such pigeon pea, cowpea, and chickpea. By growing this sorghum seed and peas on the same plot, you improve your soil and yields, and you increase varieties of food available for your household. Please note that planting two crops in that same plot makes it difficult to plant, manage, and harvest, which increases labor cost. It also makes it difficult to use machines for planting, spraying, harvesting, etc., in the case of large farms.
Commercial sorghum	This product is called "High-yielding hybrid." This sorghum seed is designed to be high yielding. It yields about 30% more compared to other improved seeds. This means that if the other improved seed gives you 10 bags, this seed will give you 13 bags, assuming you planted on a similar farm. The product is mainly for selling to traders who buy grain for food processing. You can also use it for food, but less attention has been given to cooking qualities such as taste. Please note, you will need to buy seeds every season to maintain the yield benefit. Planting seed from your harvest will reduce the yield. This product is not designed for feeding and malting.

Acronyms used in this brief

CIMMYT	International Maize and Wheat Improvement Center
EAF	Eastern African Federation
ICRISAT	International Crops Research Institute for Semi-Arid Tropics
NARES	National Agricultural Research and Extension Systems
OPV	open-pollinated variety
SPMS	Seed product market segment
TARI	Tanzania Agricultural Research Institute
TOSCI	Tanzania Official Seed Certification institute
TPP	Target Product Profile
VPCT	Video-based product concept testing

Authors

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The Market Intelligence Brief offers evidence-based insights into the potential for increased impact towards the CGIAR Impact Areas from investments in crop breeding and seed systems development. This peer-reviewed series brings together voices from diverse fields, including marketing and agribusiness, gender, plant sciences and climate change to inform debates on future priorities and investments by CGIAR, NARS, the private sector and non-governmental organizations (NGOs). This series is a collaborative effort of the CGIAR Initiative on Market Intelligence. For more information, including potential submissions, please contact lila Train, editorial assistant, at <l.train@cgiar.org>.editorial

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